

Evaluation of Clustering Algorithms

Since clustering is an unsupervised learning method, it does not have ground truth for validation. Thus, we rely on various internal and external metrics to assess the quality.

Internal Evaluation Metrics

assess the clustering quality based on the information present within the data itself, without the need of external labels

Silhouette Score

measures how similar a point is to its cluster compared to other clusters.

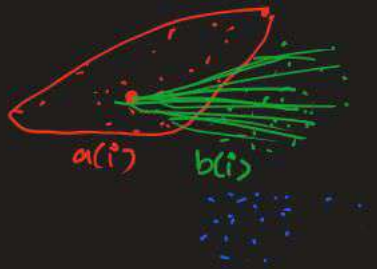
It ranges b/w -1 and +1, where a high value indicates that the points are well clustered.

for a point i ,

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

$a(i)$:- Average distance from point i to all other points in the same cluster (cohesion)

$b(i)$:- Average distance from point i to all points in the nearest cluster. (separation)



$$\frac{b(i) - a(i)}{\max(b(i) - a(i))} \uparrow$$

Calculate the average silhouette coefficient across all data points to obtain the overall silhouette score for the clustering algorithm.

eg.

Cluster A

(2, 5)

(3, 4)

(4, 6)

Cluster B

(8, 3)

(9, 2)

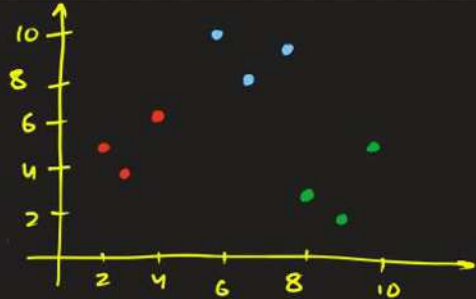
(10, 5)

Cluster C

(6, 10)

(7, 8)

(8, 9)



Calculate the silhouette score of point (2, 5) for the clustering given above.

Sol. $A1: (2, 5)$ cohesion :- $a(A1) = \frac{d(A1, A2) + d(A1, A3)}{2}$

$$= \frac{\sqrt{2} + \sqrt{5}}{2} \approx 1.825$$

Separation $b(A1) = \min [d(A1, B), d(A1, C)] \approx 6.481$

$$d(A1, B)$$

↓

$$\frac{d(A1, B1) + d(A1, B2) + d(A1, B3)}{3}$$

$$= \frac{\sqrt{40} + \sqrt{58} + \sqrt{64}}{3}$$

$$\approx 7.313$$

$$d(A1, C)$$

↓

$$\frac{d(A1, C1) + d(A1, C2) + d(A1, C3)}{3}$$

$$= \frac{\sqrt{41} + \sqrt{34} + \sqrt{52}}{3}$$

$$\approx 6.481$$

$$\text{silhouette coefficient } (A1) = \frac{b(A1) - a(A1)}{\max(b(A1), a(A1))} = \frac{6.481 - 1.825}{6.481} \approx \boxed{0.718}$$

Similarly, we can calculate silhouette coefficient for all the 9 points and calculate the average to get the silhouette score of the clustering algorithm.

Davies-Bouldin Index

is the average similarity ratio of each cluster with its most similar cluster. Lower values indicate better clustering.

For clusters i and j :

$$DB = \frac{1}{K} \sum_{i=1}^K \max_{j \neq i} \left(\frac{s_i + s_j}{d_{ij}} \right)$$

s_i : Avg. distance of points in cluster i to its centroid

d_{ij} : Distance b/w centroids of clusters i and j

Step 1: Calculate centroids for each cluster

Step 2: Calculate the intra-cluster distance (S_i)

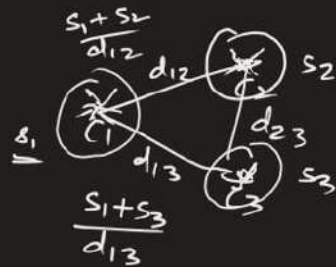
For each cluster i , compute the average distance of all points in the cluster from the centroid of the cluster

Step 3: Calculate the inter-cluster distance (d_{ij})

For each pair of clusters i and j , d_{ij} = distance b/w the centroids of i and j .

Step 4: For each cluster i , compute maximum value of

$$\frac{S_i + S_j}{d_{ij}} \text{ for all } j \neq i$$



Step 5: Finally, average these max.

values to get the Davies-Bouldin

index for the clustering algorithm

eg.

Cluster A

(2, 5)

(3, 4)

(4, 6)

Cluster B

(8, 3)

(9, 2)

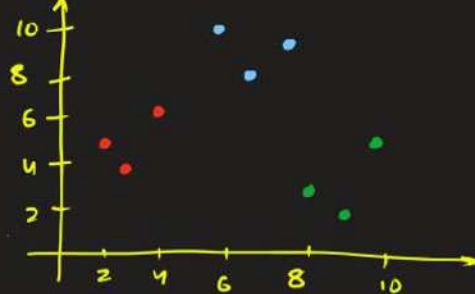
(10, 5)

Cluster C

(6, 10)

(7, 8)

(8, 9)



Calculate the Davies-Bouldin index for the clustering given above.

Sol.

Step 1:

$$C_A = \left(\frac{2+3+4}{3}, \frac{5+4+6}{3} \right) = (3, 5)$$

$$C_B = \left(\frac{8+9+10}{3}, \frac{3+2+5}{3} \right) = (9, 3.33)$$

$$C_C = \left(\frac{6+7+8}{3}, \frac{10+8+9}{3} \right) = (7, 9)$$

$$\text{Step 2: } S_A = \frac{d((2, 5), (3, 5)) + d((3, 4), (3, 5)) + d((4, 6), (3, 5))}{3}$$

$$= \frac{1 + 1 + \sqrt{2}}{3} \approx 1.138$$

$$S_B = \frac{1.053 + 1.33 + 1.9465}{3} \approx 1.443$$

$$S_C = \frac{\sqrt{2} + 1 + 1}{3} \approx 1.138$$

Step 3 $d_{AB} \approx 6.23$, $d_{BC} \approx 6.012$, $d_{AC} = 5.66$

Step 4 $dB(A) = \max\left(\frac{1.138 + 1.443}{6.23}, \frac{1.138 + 1.138}{5.66}\right)$
 $= \max(0.414, 0.402) = 0.414$

$$dB(B) = \max\left(0.414, \frac{1.443 + 1.138}{6.012}\right)$$

$$= \max(0.414, 0.429) = 0.429$$

$$dB(C) = \max(0.402, 0.429) = 0.429$$

Step 5: $DB \text{ index} = \frac{0.414 + 0.429 + 0.429}{3} \approx \boxed{0.424}$